

UCE855: DESIGN OF REINFORCED EARTH STRUCTURES				
	L	T	P	Cr
	3	1	0	3.5
Course Objective: To learn (i) basic concepts and applications of various types of soil reinforcement, and (ii) design of reinforced earth structures				
Detailed Syllabus 1. Introduction to Reinforced Earth Structures Historical Background and Evolution, Basic Principles and Concepts of Reinforced Earth, Mechanism of Reinforcement and Load Transfer, Advantages and Applications of Reinforced Earth, Comparison with Conventional Retaining Structures 2. Materials Used in Reinforced Earth Structures Types of Reinforcements: Geosynthetics, Metallic Strips, Geogrids, Geocells, Geomembranes, Properties and Selection Criteria for Reinforcement Materials, Soil-Reinforcement Interaction, Testing and Quality Control of Reinforcement Materials, Durability and Long-Term Performance of Reinforcements 3. Design and Analysis of Mechanically Stabilized Earth (MSE) Walls Types of Reinforced Earth Retaining Walls, Failure Mechanisms: Internal, External, and Compound Failures, Design Methods: Limit Equilibrium and Limit State Approaches, Seismic Considerations in Reinforced Earth Walls, Drainage and Facing Considerations, Case Studies of Reinforced Retaining Walls, Lessons from Failures in Reinforced Earth Structures 4. Design of Reinforced Slopes and Landslide Mitigation Types of slopes and their stability, Reinforcement Methods for Slope Stabilization, Design of Reinforced Slopes using Geosynthetics and Soil Nailing, Retaining systems involving breast walls, gabion walls, Landslide Control Measures and Case Studies 5. Codal provisions Review of related codes like SP102, IS14458, IS14680, Applicability for different situations.				
Experimental Project/assignment/Micro Project Numerical modelling and simulation of Reinforced Earth Walls and Slopes using RS3, Slide3				
Course Learning Objectives (CLO) The students will be able to: 1. Learn the concept of reinforced earth and factors affecting same Analyze and design plate girders 2. Design reinforced earth retaining wall 3. Application of reinforcement to stabilize slope and evaluate stability 4. Prepare and analyse computer aided numerical model of retaining wall and slope				
Text Books: 1. Bowels J.E., Foundation Analysis and Design, McGraw Hill Book Company (1997), 2. Shukla, S.K., Yin, Jian-Hua, Fundamentals of Geosynthetic Engineering, Taylor & Francis (2006). 3. Koerner, R., Designing with Geosynthetics, 6th Ed. Prentice Hall (2005), 4. Saran, S., Reinforced Soil and Its Engineering Applications, I.K. international (2005).				

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weights (%)
1.	MST	30
2.	EST	45
3.	Sessionals (May include Assignments/Projects/Quizzes/Tutorials/Lab Evaluations)	25